Ian Lingo

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Education

Texas State University — San Marcos, TX
B.S. in Computer Science — *Graduated: May 2026*

- **GPA**: 3.58
- **Minors**: Applied Mathematics, Mass Communication
- **Honors**: Cum Laude, IEEE-HKN (Eta Kappa Nu), Texas State Honors College
- **Coursework**: Data Structures & Algorithms, Database Systems, Linear Algebra, Computer Architecture, Digital Logic, Computer Vision, Object-Oriented Programming, Operating Systems, Systems Security, Software Engineering

San Jose State University - Open University — San Jose, CA
Advanced Python Programming — *Jun. 2026 – Aug. 2026*

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript, SQL
Embedded Systems: Jetson Orin Nano, Teensy, Arduino, UART, USB Serial, PWM, I2C, GPIO, Encoders, FPGA Basics
Frameworks/Tools: React, Next.js, Flask, Spring Boot, Gradle, JUnit, REST APIs, ROS2, Docker, YOLO, OpenCV
Cloud/DevOps: Git, GitHub, GitHub Actions, CI/CD, AWS, OCI, Jira, Bitbucket, Vercel, Firebase
Data: PostgreSQL, SQL Server, BigQuery, Tableau, Pandas, NumPy, Matplotlib, Jupyter Notebook
AI/Developer Tools: Claude Code, GitHub Copilot, ChatGPT, Google Gemini
Operating Systems: Linux (Ubuntu), Windows, macOS, iOS
Certificates: Google Data Analytics, Oracle Cloud (OCI) Foundations, AWS Certified Developer - Associate (In Progress)

Projects

MARS Robot | *ROS2, Embedded Systems, Python, C++, YOLO, ML, Computer Vision* — Sep. 2025 – May 2026

- Directed a **17-member** IEEE RAS engineering team in Agile Scrum to develop an indoor service robot platform using NVIDIA Jetson Orin Nano, ZED 2i, RPLidar A1, ToF sensors, CAD, and embedded motor control.
- Built and deployed a **YOLOv11 Nano ML & CV pipeline** from a **229-image annotated Roboflow dataset** into a ZED 2i camera pipeline for real-time industrial object detection.
- Developed **C++ PWM motor-control firmware** on Teensy 4.1 and integrated it with **Python ROS2 software** for perception, human tracking, and Xbox teleoperation through Jetson-to-Teensy communication.
- Integrated, tested, and debugged hardware/software interfaces across embedded control, sensors, and ROS2 systems.

Chess Game Application | *Java, Spring Boot, PostgreSQL, Docker, AWS EC2, JUnit 5* — Jul. 2024 – Present

- Engineered a **full-stack** Java chess app (Swing client + Spring Boot REST API) with a multi-module Gradle build and a from-scratch **AI opponent** using negamax, alpha-beta pruning, and quiescence search.
- Built an Elo-rated leaderboard with a Spring Boot REST API backed by PostgreSQL; containerized with Docker, **deployed to AWS EC2**, covered by **20 JUnit 5 tests** using Mockito and Testcontainers in a **GitHub Actions CI** pipeline.

IEEE TXST Website Platform | *Next.js, TypeScript, Tailwind CSS, Vercel, Cloudflare* — Dec. 2025 – May 2026

- Led a **7-member** student development team to design, build, and deploy a production web platform for IEEE Texas State University using Next.js 15 (App Router), replacing a legacy WordPress site serving **100+ monthly users**.
- Developed responsive UI with Tailwind CSS, Framer Motion, and Swiper; configured Cloudflare DNS, custom routing, automated **Vercel CI/CD deployment**, and implemented SEO/performance optimizations including metadata.

Experience

Geek Squad Advanced Repair Technician — Jun. 2026 – Present
Best Buy — *San Jose, CA*

- Diagnose, troubleshoot, and repair **PCs, Apple devices, and Windows/macOS/Linux systems**; document **Apple GSX** repair workflows, build computers, and explain technical solutions to clients as an **Apple-certified technician**.

Director, Vice President, Treasurer, and Co-Founder — Aug. 2024 – May 2026
Boko Hackathon, IEEE, IEEE-HKN, and IEEE RAS Chapter — *San Marcos, TX*

- Developed **C++ mecanum movement-control software** using PWM motor control, encoder feedback, USB serial communication, and I2C ToF sensing; built **Python AprilTag computer-vision code** on a **5-member robotics team** that placed **2nd out of 25 universities** at the IEEE Region 5 Robotics Competition.
- Co-founded IEEE Robotics & Automation Society (RAS), scaling to **100+ members** and directing a **17-member** team.
- Co-directed an AI and cybersecurity hackathon with **100+ participants**, served as a technical code judge, and raised **$12,000+** from Google, Dell, BrainTrust, UFCU, IEEE, and Charles Schwab via sponsorship outreach.

Team Lead and Learning Ambassador (Operations Associate) — Jan. 2021 – Jul. 2025
Amazon — *San Antonio, TX and Austin, TX*

- Led and trained a **10+ member** outbound operations team, monitoring supply-chain throughput of **10,000+ packages daily** and implementing process improvements that reduced truck dispatch time by **50%**.